Neural Consensus Protocol (NCP)

Mathematical Foundations and Algorithm Specification

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1. Formal Definitions

Definition 1 (Reasoning Chain). A reasoning chain C is a tuple (id, a, v, S, σ) where: id ∈ H is a unique identifier, a ∈ A is the owning agent, v ∈ N■ is the version, S = (s■, s■, ..., s■) is an ordered sequence of steps, σ ∈ {active, broken, archived} is the chain status.

Definition 2 (Step). A step s■ is a tuple (i, c■, τ■, n■) where: i ∈ N■ is the index, c■ is the assertion content, τ■ ∈ {active, collapsed, fortified} is the status, n■ ∈ N is the attack count.

Definition 3 (Attack). An attack X is a tuple (a_src, C_tgt, i_tgt, r, ω) where: a_src is the attacker, C_tgt is the target chain, i_tgt is the target step, r is the reason, ω ∈ {0,1} is the outcome.

Definition 4 (Address). Every step has a unique address:

```
addr(s) = agent.v{version}.Step{index} e.g., GPT-5.4.v2.Step3
```

2. Entropy Calculations

2.1 10-Dimensional Entropy Vector

The system state is characterized by:

```
H(t) = (H_sem, H_cau, H_bnd, H_tmp, H_dep, H_div, H_inf, H_prp,
H_evi, H_imp)
```

where each H■ ∈ [0, 1] measures a distinct type of uncertainty.

2.2 Divergence Entropy (Stance-Based)

For each active chain, define a stance vector v_a ∈ {-1, 0, +1}^d:

```
v_{a,j} = +1 if positive signals dominate, -1 if negative, 0
otherwise
```

Stance agreement between agents a, b:

```
agree(a,b) = (1/|D|) × Σ_{j∈D} match(v_{a,j}, v_{b,j})
```

where match = 1 if equal, 0.5 if one is 0, 0 otherwise.
Divergence entropy:

```
H_div = 1 - (2 / |A|(|A|-1)) × Σ_{a<b} agree(a,b)
```

2.3 Total Entropy with Impact Amplification

Theorem 1. The total entropy is amplified by impact:

```
H_total = min(1, H_mean × (1 + 0.5 × H_imp))
```

Amplification range: 1.0× (no impact) to 1.5× (max impact).

3. UCB1 Attack Scheduling

3.1 Multi-Armed Bandit Formulation

Definition 5 (UCB1 Score). For step s■:

```
UCB(s■) = c■/n■ + √2 × √(ln(N)/n■) + 0.1 × ln(1 + Δt■)
```

where c■ = collapse count, n■ = attack count, N = total attacks, Δt■ = rounds since last attack

> **Property 1 (Exploration Guarantee).** If n■ = 0, then UCB(s■) = +∞. Every step is attacked at least once before any step is attacked twice.

Theorem 2 (Coverage Bound). With |A| agents and |S| total steps, after r rounds:

```
coverage(r) ≥ min(1, r × |A| / |S|)
```

With 30 agents and 210 steps: full coverage in ■210/30■ = 7 rounds.

4. Decentralized Parallel Voting

Definition 6. Voting set: V = A \ {a_src} (all except attacker)

```
Each voter independently: vote(v) ∈ {collapsed, defended}
```

Majority Decision:

```
ω = 1 (collapsed) if |{v: vote(v)=collapsed}| > |{v:
vote(v)=defended}|, else 0
```

> **Property 2 (No Privileged Node).** No judge, no arbiter. Attacker cannot vote.

> **Property 3 (Parallel Execution).** latency(voting) = max_{v∈V} latency(v) = O(1) w.r.t. |V|

Theorem 3 (Byzantine Fault Tolerance). With n voters, tolerates ■(n-1)/2■ Byzantine nodes.
Proof: Byzantine voter contributes at most 1 wrong vote. Honest majority n-f > n/2 when f < n/2. ■

5. Convergence Theory

Definition 7 (Verified Convergence). System converges when:

```
coverage(t) > θ_c AND verified_ratio(t) > θ_v AND t ≥ t_min
```

Default: θ_c = 0.5, θ_v = 0.4, t_min = 2
Theorem 4 (Convergence Speed). Expected rounds:

```
E[T] ≤ ■θ_c × |S| / |A|■ + 1
```

Corollary: 30 agents, 49 steps, θ_c=0.5: E[T] ≤ 2 rounds.
This proves: **more agents** → **faster convergence**.

6. ELO Reputation System

After attack between attacker a and defender d:

$$E_a = 1 / (1 + 10^{((R_d - R_a)/400)})$$

$$R_a' = R_a + K \times (\omega - E_a)$$

$$R_d' = R_d + K \times ((1-\omega) - E_d)$$

$K = 32$, $\omega = 1$ if attack succeeds, 0 otherwise.

Vote weight mapping:

$$w(a) = \mathrm{clip}((R_a - 800)/400 + 0.5, 0.5, 2.0)$$

7. Consensus Persistence with Decay

Confidence decay function:

$$\phi(t) = \max(\phi_{floor}, 1.0 - \lambda(t-t■) + \mu \times n_{ref}(t))$$

$\lambda = 0.005$/day, $\phi_{floor} = 0.3$, $\mu = 0.02$/reference

$$\text{Challenge penalty: } \phi' = \phi - 0.15$$

Revocation propagation: if $\phi' \leq 0.1$, all dependents lose 0.2

7.1 Cross-Session α Calculation

$$\alpha(c) = \Sigma_{\{a \in A\}} w(a) \times 1[c \in C_a] / \Sigma_{\{a \in A\}} w(a)$$

Convergence threshold: $\alpha \geq 0.65 \rightarrow$ TRUE_CONSENSUS

8. Protocol Invariants

I1: $\forall C$: $v_C \geq 1$ (version is positive)
I2: $\forall s$: $n_s \geq 0$ (attack count non-negative)
I3: τ_s = fortified ■ $n_s \geq 2$ (fortification requires 2+ survived attacks)
I4: σ_C = broken ■ $\exists s \in C.S$: τ_s = collapsed
I5: REPAIR(C) ■ $v_{\{C'\}} = v_C + 1$ (repair increments version)
I6: σ_C = archived ■ C cannot be attacked
I7: vote(a_src, X) = $\perp$ (attacker cannot vote on own attack)
I8: $H_{total} \in [0, 1]$ (entropy is bounded)
I9: $\alpha(c) \in [0, 1]$ (consensus degree is bounded)
I10: $\phi(t) \geq \phi_{floor}$ unless revoked

9. Complexity Summary

Component	Per-Round Complexity	Bottleneck
Chain parsing	O(|A| × k)	Regex
UCB1 scoring	O(|S|)	Step traversal
Attack routing	O(|A| × |S|)	Display build
Parallel voting	O(1) latency	Slowest model
Entropy	O(|A| × k)	Stance extraction
α extraction	O(1) call	Single model
Consensus store	O(|C_store|)	Keyword match

True bottleneck: LLM inference latency (~10-30s). All algorithmic overhead is sub-second.

10. Scaling Law

Agents	Attacks/Round	Coverage/Round	Expected Convergence
3	3	~15%	4-6 rounds
7	7	~35%	2-3 rounds
15	15	~70%	1-2 rounds
30	30	~95%	1 round

The fundamental scaling law: More agents → more parallel attacks → higher coverage per round → faster convergence. This is the opposite of traditional multi-agent systems.